

# Tarun Mittal

## Senior Software Engineer

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### SUMMARY

- Senior Software Engineer with **5 years of experience** designing and owning cloud-native, high-throughput distributed systems.
- Expertise in **Python and Go microservices**, event-driven architectures, real-time and batch data processing, **Agentic AI and RAG systems**.
- Proven track record shipping **production systems**, leading backend migrations, and improving reliability, performance, and scalability.

### SKILLS

- Languages:** Python, Go (Golang), SQL, C, C++.
- Backend & APIs:** FastAPI, Gin, Django, SQLAlchemy, Pydantic, REST, gRPC, Celery, asyncio, Design Patterns.
- AI & Agents:** LLM agents (OpenAI/Gemini/Claude), LangGraph, LangChain, MCP, RAG, pgvector.
- Data & Streaming:** Kafka, Flink, Spark, ClickHouse, BigQuery, PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, DynamoDB, S3.
- Cloud & Infra:** GCP, AWS, Docker, Kubernetes, Linux, CI/CD, Bazel, Observability (ELK, New Relic, Datadog, PostHog, LangSmith).
- Core:** Distributed Systems, Microservices, System Design (HLD/LLD), Fault Tolerance, Concurrency, High Availability, Caching, Sharding, Performance Tuning, Mentorship.

### PROFESSIONAL EXPERIENCE

Senior Software Engineer IMPACT ANALYTICS, Bangalore May 2026 – Present

#### Agentic AssortSmart (AI Powered Retail Merchandise Planning Platform)

- Architecting Agentic AssortSmart: AI-drafted store clusters and plans for planner approval under gated write-back.
- Developing the **Cluster Recommendation Copilot (FastAPI, LangGraph, MCP)** that owns chat and calls a **Go doing layer** as tools, batch evaluating **20 to 100 configurations per plan vs 1 manual attempt**, targeting turnaround **under 1 hour** (from days).
- Enforcing **read only agent tools** and **3 approval gates**, targeting failures under 2% from 8.5%, **100% reproducible** clusters, and **p95 probes under 500ms** vs 1 to 20s on BigQuery.
- Designing **Go (Gin) microservices** for manual REST and agent tool paths (Hindsight, Clustering, Strategy) with JWT auth, plus **Datadog, LangSmith**, and **PostHog** under a shared OTEL trace id.
- Building Agentic AssortSmart on **ClickHouse/GCS** as the end-to-end planning store (insert-only versioned writes): POC cut **250M-row** pivot grids from **189s to 12.3s** (~15x) and a line-plan aggregate avoided materializing **12B** store-week rows (**100 to 450x**).
- Tech Used:** Python, FastAPI, LangGraph, MCP, Go, Gin, ClickHouse, BigQuery, GCS, Datadog, LangSmith, PostHog, GCP, Docker.

Software Development Engineer 2 UBER (via EPAM Systems), Bangalore July 2024 – May 2026

#### Financial Risk Management (FRM) Scoping Platform (Uber Finance)

- Owned Uber's **FRM Risk Scoping platform (FastAPI, MySQL, React)** across **8 screens** and **30+ REST APIs**, automating quarterly scoping at **\$340M group materiality** and targeting a **70% cut** in manual reconciliation (~2 weeks to ~3–4 days).
- Designed the software architecture for an **11 table SQLAlchemy 2.0 MySQL schema** with layered handler, service, and repository boundaries, auto flagging material items across **55 line items** and **14 entities**.
- Owned Sheets to **MySQL v2 recon migration** (18 files), validating HFM extracts against public **10-Q** filings and fixing a latent column aliasing bug.
- Led 3 engineers**, owning design reviews, API contracts, and CI quality gates for releases feeding PwC audit work papers.
- Tech Used:** Python, FastAPI, SQLAlchemy 2.0, Pydantic, MySQL, Bazel, Docker.

#### Menu Ingestion & Automation Platform (Uber Eats)

- Engineered menu ingestion with **Selenium scrapers on GCP** publishing to **Kafka** (~200–500 peak events/sec), with **Flink** normalize/dedupe and **Spark** backfills, for **30K+ menus/month**, cutting onboarding 24h to 2h and saving **\$600K+ annually**.
- Shipped a **RAG and Gemini 2.5 Pro** pipeline with supervised fine tuning, hitting **98% fidelity** and **100% schema consistency** (offline eval).
- Defeated anti bot defenses with IP rotation and dynamic proxy pools, increasing successful ingestions **95%**.
- Automated **ANZ driver document compliance** to 99.9%, saving 20 hours/week of manual verification.
- Tech Used:** Python, Selenium, Kafka, Flink, Spark, Gemini, RAG, SFT, GCP, Docker.

Software Development Engineer 2 MASTERS INDIA, Noida December 2022 – June 2024

#### GST Compliance & E-Invoicing Platform

- Owned the PHP monolith to **FastAPI microservices** strangler migration and **mentored 2 engineers**, cutting p95 latency from **1.2s to 300ms** for 1,500+ clients.
- Scaled bulk e-invoicing to **100K+ per import** and **1M+ daily** (~12 TPS avg, 100+ peak) via async Kafka and PostgreSQL quarter sharding, lifting throughput from **700 to 4,000 requests/min** (~67 RPS).
- Built fault tolerant bulk paths with **idempotency keys, retries, and DLQ replay**, plus Redis caching that cut redundant database reads **30%**.
- Established **ELK and New Relic** on-call alerting, cutting incident triage 70%, and raised coverage from **35% to 82%** with 98% deployment success.
- Tech Used:** Python, FastAPI, Kafka, PostgreSQL, MongoDB, Redis, Elasticsearch, Celery, Docker, ELK, New Relic, AWS.

Software Development Engineer GEEKSFORGEEKS, Noida August 2021 – November 2022

- Migrated the doubt support backend from PHP to **Django (Python)**, improving reliability for **10K+ daily queries** and **10x contest spikes**.
- Designed voting and pinning APIs on MySQL, MongoDB, Redis, and Elasticsearch, lifting **premium subscriptions 15 to 20%**.
- Launched an influencer analytics dashboard and cron pipelines for video processing, raising **course sales 30%** and operational efficiency 70%.

### EDUCATION

B.Tech (Information Technology) Institute of Engineering & Technology, Lucknow July 2017 – June 2021

### ACHIEVEMENTS & CERTIFICATIONS

- Ranked **2260 / 37,000+** in the Google Code Jam Qualifier 2021 and finalist (top 3 nationally) at the **Smart India Hackathon 2020 (MHRD)**.
- Participated in the **Global AI Hackathon** at EPAM Systems.
- Certifications:** [HackerRank Problem Solving](#), [LangChain Academy](#).